A FINITE GROUP ACTING FAITHFULLY ON A FIELD GIVES A GALOIS EXTENSION OF DEGREE THE ORDER OF THE GROUP OVER THE FIXED FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The author flags the G-invariance of the orbit polynomial as 'the key to the whole lemma' (source line 2956) and carries the permutation computation in full (source lines 2958-2962).

Editorial note. The label lemma-galois-over-fixed-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2935-2941: definition of the fixed field K^G
- Same chapter: lemma-finite-Galois (finite extension Galois iff |Aut(E/F)| = [E:F], stated and proved at source lines 2880-2894), lemma-recognize-separable (a divisor of a separable polynomial is separable), lemma-primitive-element (finite separable extensions are simple), lemma-normal-and-automorphisms (the automorphism bound inequality)

Source reference. Original reference 09I1: sources/stacks-fields/pinned-original-source.tex, source lines 2881–2881. Prerequisite pointer invoked at 2973; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HS: sources/stacks-fields/pinned-original-source.tex, source lines 2211–2211. Prerequisite pointer invoked at 2997; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 030N: sources/stacks-fields/pinned-original-source.tex, source lines 2518–2518. Prerequisite pointer invoked at 2991; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09H3: sources/stacks-fields/pinned-original-source.tex, source lines 1424–1424. Prerequisite pointer invoked at 2969; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let G be a group acting on a field K (by field automorphisms). We will often use the notation

$$K^G = \{x \in K \mid \sigma(x) = x \ \forall \sigma \in G\}$$

and we will call this the *fixed field* for the action of G on K.

Lemma 0.1. *Let K be a field. Let G be a finite group acting faithfully on K. Then the extension K/K^G is Galois, we have $[K : K^G] = |G|$, and the Galois group of the extension is G.*

Proof. Given $\alpha \in K$ consider the orbit $G \cdot \alpha \subset K$ of α under the group action. Consider the polynomial

$$P = \prod\nolimits_{\beta \in G \cdot \alpha} (x - \beta) \in K[x]$$

The key to the whole lemma is that this polynomial is invariant under the action of G and hence has coefficients in K^G. Namely, for $\tau \in G$ we have

$$P^\tau = \prod\nolimits_{\beta \in G \cdot \alpha} (x - \tau(\beta)) = \prod\nolimits_{\beta \in G \cdot \alpha} (x - \beta) = P$$

because the map $\beta \mapsto \tau(\beta)$ is a permutation of the orbit $G \cdot \alpha$. Thus $P \in K^G[x]$. Since also $P(\alpha) = 0$ as α is an element of its orbit we conclude that the extension K/K^G is algebraic. Moreover, the minimal polynomial Q of α over K^G divides the polynomial P just constructed. Hence Q is separable (by Lemma 09H3 for example) and we conclude that K/K^G is separable. Thus K/K^G is Galois. To finish the proof it suffices to show that $[K : K^G] = |G|$ since then G will be the Galois group by Lemma 09I1.

Pick finitely many elements $\alpha_i \in K$, $i = 1, \ldots, n$ such that $\sigma(\alpha_i) = \alpha_i$ for $i = 1, \ldots, n$ implies σ is the neutral element of G. Set

$$L = K^G(\{\sigma(\alpha_i); 1 \leq i \leq n, \sigma \in G\}) \subset K$$

and observe that the action of G on K induces an action of G on L. We will show that L has degree $|G|$ over K^G. This will finish the proof, since if $L \subset K$ is proper, then we can add an element $\alpha \in K$, $\alpha \notin L$ to our list of elements $\alpha_1, \ldots, \alpha_n$ without increasing L which is absurd. This reduces us to the case that K/K^G is finite which is treated in the next paragraph.

Assume K/K^G is finite. By Lemma 030N we can find $\alpha \in K$ such that $K = K^G(\alpha)$. By the construction in the first paragraph of this proof we see that α has degree at most $|G|$ over K. However, the degree cannot be less than $|G|$ as G acts faithfully on $K^G(\alpha) = L$ by construction and the inequality of Lemma 09HS. $\square$